Aaron Chou

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EDUCATION

Carnegie Mellon University (CMU), *M.S. in Robotic Systems Development (MRSD)* — Pittsburgh, PA

- **Coursework:** Introduction to Robot Learning — Aug. 2026 – May 2028

National Taiwan University (NTU), *B.S. in Mechanical Engineering* — Taipei, Taiwan

- Last 60 GPA: 4.03/4.3 | CGPA: 3.80/4.3 — Sept. 2018 – Jan. 2023
- **Coursework:** Automatic Control, Digital Control System, Kinematics, Dynamics, Computer Programming

Aoyama Gakuin University (AGU), *Exchange Program* — Tokyo, Japan, Fall 2022

PUBLICATIONS

[1] **Y.L. Chou**, L.C. Wang, Y.S. Luo, Y.T. Liu, J.L. Li. *VL-N3RD-Bench: Benchmarking Vision-Language Navigation with 3D Gaussian Splatting Reconstruction for Deployment*, ICRA Workshop: VLA Pipelines for Real Robots, 2026.
[2] Y.S. Luo, L.C. Wang, H. Mandala, **Y.L. Chou**, G.H.G. Christmann, C.Y. Lee, W.C. Chen, et al. *Feasibility-Guided Planning over Multi-Specialized Locomotion Policies*, IEEE Int. Conf. Robot. Autom. (ICRA), 2026. (link)

EXPERIENCE

Stealth-Mode Startup — Mountain View, CA
Robotics Research Intern — UMI, Diffusion Policy — May 2026 – Aug. 2026

Inventec Corporation, AI Center, Robotics — Taipei, Taiwan
Robotics Research Engineer — Quadruped Robots — Mar. 2025 – May 2026

- **Vision-Language Navigation (VLN):** Integrated and evaluated a pretrained VLA model for real-world navigation, and built an **IsaacLab** benchmark to analyze how **3D Gaussian Splatting** (3DGS) reconstruction quality affects navigation performance.
- **3DGS Benchmarking:** Benchmarked SoTA 3DGS pipelines using standard image-quality metrics and runtime/memory analysis.
- **Learning-Based Control:** Developed low-level control infrastructure with LCM communication, custom motor interfaces, calibration routines, and IMU integration for quadruped platforms with no official API, deploying **IsaacGym**-trained policies (link).
- **Terrain-Aware Navigation:** Trained a 15 cm gap-traversal policy, implemented a feasibility-guided planning framework for hybrid terrain skill selection, conducted real-world experiments, and contributed to an ICRA 2026 submission [2].
- **SLAM Optimization:** Improved single-LiDAR SLAM by **dual-LiDAR fusion** (link) in ICRA 2025 QRC, achieving a **65%** reduction in mapping-to-navigation time and enabling rapid terrain reconstruction in dynamic environments.

Flytech Technology Co. Ltd. — Taipei, Taiwan
Robotics Engineer — Autonomous Mobile Robots — Mar. 2024 – Jan. 2025

- Engineered a precise docking system (demo) by integrating motion control, localization, and a customized rack tracker, achieving navigation accuracy of **±1–1.5 cm** to minimize redocking attempts in production deployments.
- Developed and maintained **10+ ROS packages** and delivered a patrol demo showcased at Computex 2024.

Chien Kuo High School, Robotics team, FRC#8020 CyberpunK — Taipei, Taiwan
Youth Mentor — Mechanical Design — Feb. 2022 – Jul. 2023

- Mentored 30+ students for robot design, CAD/CAM, CNC operation, and testing of the competition-ready robot (demo).

SKILLS

Robotics: IsaacSim/Lab, ROS/ROS2 (Nav2, MoveIt2, FAST-LIO), 3D Reconstruction (3DGS), Sensor Fusion, SLAM
Hardware & Software: Unitree Go1/A1, Flexiv Rizon, Livox LiDAR, RealSense, IMU, RTX 5090; Docker, Git, Blender, OnShape
Programming: Python, C++, Shell Scripting (Bash), LaTeX

HONORS & COMPETITIONS

2025 ICRA Quadruped Robot Challenge (QRC) — Participation Certificate, Autonomous (demo) — Atlanta, GA
2022 FIRST Robotics Competition (FRC) Sacramento Regional, Finalist (Team Mentor) — Sacramento, CA
2020 Dean's List Award (top 5% of the class in the semester) — Taipei, Taiwan